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RESEARCH ARTICLE

Question Answering on Lengthy Arabic Narrative Books

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ABSTRACT Despite the growing body of Arabic question-answering (QA) resources, most existing datasets focus on factoid questions or moderately sized texts, leaving a significant gap in addressing the complexities of lengthy narrative books. In this paper, we introduce AQANB, a new Arabic QA dataset specifically curated for extended narrative texts. This dataset enables the development and evaluation of QA systems capable of handling intricate narrative structures and broader contextual dependencies. Answering questions over long narratives is challenging due to the inherent token limitations of large language models, which restrict the amount of text processed in a single pass. Moreover, processing a larger number of tokens significantly increases computational costs. Furthermore, processing only a selected set of relevant passages requires question-passage annotation to train the retriever. To tackle this, we propose an annotation-free retriever training framework. Instead of relying on manually annotated data, our approach leverages large language models to generate synthetic question-passage pairs that probe deeper semantic content. Crucially, we introduce a novel narrative coherence-based scoring mechanism that assigns high relevance not only to the most pertinent passage but also to the passages surrounding it, thereby capturing continuity and context rather than treating adjacent passages as purely negative samples. This design improves retrieval accuracy for narrative-style content where crucial information may span multiple, interconnected sections. Our experiments on AQANB demonstrate the effectiveness of the proposed approach. We achieve relative improvements of 21.7% and 11.7% over OpenAI's small and large embedding models, respectively, and outperform the open-source multilingual Contriever by 29.6%. Furthermore, adding our retriever as a pre-stage to the common OpenAI GPT-4o reduced the cost by 98 times and achieved better performance compared to processing the entire book with GPT-4o.

INDEX TERMS Question answering for extended narratives, comprehension of lengthy books, unsupervised retriever training, arabic narrative QA.

I. INTRODUCTION

Arabic novels are lengthy stories that contain complex events, well-developed characters, and intricate interactions. The main challenges in processing these texts are understanding the large number of pages and capturing the relationships between events and characters across the story. Narrative QA systems can help readers by allowing them to ask questions about any element of the story, including characters, events, or interactions, which enables better comprehension and analysis.

Factoid QA [1], [2], [3] often requires concise answers derived from isolated passages. In contrast, narrative QA demands systems capable of understanding logical connections, interactions between multiple paragraphs, and the broader context necessary for constructing comprehensive answers [4]. For example, passages that may initially appear unrelated to the question could be crucial for maintaining

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narrative coherence and constructing a full sequence of events. These complexities make QA systems designed for narrative books essential for real-world applications involving lengthy and interconnected texts.

To our knowledge, the current literature still lacks the resources and methods required for Arabic narrative QA. Existing Arabic QA datasets such as ArabicaQA [1], Arabic-SQuAD [3], ARCD [3], HAQA, and QUQA [2] focus mainly on factoid questions and short texts, which leaves a clear gap in addressing long Arabic narrative books that require deeper contextual and cross-paragraph understanding. Although a few recent studies [5] have explored narrative QA in Arabic, these efforts remain limited to short documents, depend on annotated question–passage pairs, or use retrieval methods that emphasize direct relevance while overlooking narrative coherence.

This study addresses these gaps by introducing AQANB,[1] an Arabic dataset specifically designed to evaluate QA systems on long narrative books. In addition, we propose an annotation-free method that leverages large language models (LLMs) to generate challenging question–passage pairs without requiring manual labeling. Unlike existing retrievers [1], [5], which rely solely on question–passage relevance, our approach trains the retriever to identify underlying semantic events and prioritize passages that maintain narrative continuity. This design enhances retrieval accuracy and enables more reliable handling of interconnected narrative text.

The NarrativeQA dataset [6] is widely recognized for its focus on lengthy narrative books in English. However, the recently introduced Arabic NarrativeQA dataset [5] is limited to books of up to 2,000 tokens, which restricts its usefulness for modeling long-context understanding in Arabic. This shortage of resources hinders the development of systems capable of processing extensive Arabic texts. Therefore, we introduce the AQANB dataset, specifically curated for lengthy narrative books, with an average document length of 150,000 tokens, making it suitable for evaluating Arabic narrative QA models.

Addressing QA tasks on extensive texts, such as lengthy narrative books, requires strategies that balance computational efficiency with the ability to generate accurate responses. One approach involves expanding the model's context window to process longer text sequences. However, this method presents significant challenges: it is computationally intensive, requiring the processing of all tokens in the input [7], and can lead to a reduction in performance [8], [9]. A critical issue, highlighted by [8], is that language models underperform when key information appears in the middle of long inputs, compared to when it appears at the beginning. This highlights the inherent difficulty of maintaining focus on relevant content across extended contexts and underscores the need for strategies that reposition pertinent information to improve performance in narrative QA tasks.

Alternatively, Retrieval-Augmented Generation (RAG) provides an effective solution for handling extensive texts. This method uses efficient retrievers to identify and extract the passages most relevant to the input question. By focusing on pertinent passages and feeding them into a language model with a limited context window, RAG avoids the inefficiencies of processing entire texts. This approach not only enhances computational efficiency but also addresses challenges in maintaining focus on critical information, especially in the middle of long narratives. However, RAG systems face two key challenges: (1) Training effective retrievers often requires large datasets of question-passage pairs, which are not always readily available [10]; and (2) Existing retrieval methods primarily focus on question-passage relevance, often resulting in fragmented retrieval. This fragmentation occurs when passages crucial for maintaining event continuity are excluded simply because they lack direct relevance to the question.

In this paper, we introduce a novel method to train retrievers efficiently without requiring annotated data, addressing key challenges in narrative QA tasks. Unlike traditional approaches that focus solely on retrieving passages semantically relevant to the question, our method prioritizes passages surrounding the relevant one. This design is based on the assumption that adjacent passages complement central narrative events and capture interactions across consecutive segments, which is critical for maintaining narrative coherence.

To train the retriever without the need for costly manual annotations, we leverage LLMs, to construct synthetic question-passage pairs. Our approach follows a two-stage strategy: first, the LLM extracts the main events from a passage, and second, these events are used as input to generate challenging questions. This method ensures that the generated questions are semantically tied to the passage's events while avoiding direct token overlap. Consequently, our retriever is trained to match the underlying semantic events rather than relying on token overlap, making the model more robust to linguistic variation.

In this research, we aim to fulfill three primary objectives:

1) Introducing the Arabic QA dataset, specifically tailored for lengthy narrative books.
2) Proposing an annotation-free framework to train the retriever based on semantic relevance between questions and passages without relying on lexical overlap, enabling robust retrieval without manual annotations.
3) Proposing a narrative coherence-based scoring mechanism for pseudo-labeling, enabling the retriever to prioritize contiguous passages and preserve narrative continuity.

Our contribution in this research is three-fold:

1) **A new Arabic benchmark for long-form narrative QA:** We introduce the AQANB dataset, specifically

[1]The dataset, along with all scripts used for training the models, is publicly available at https://github.com/MohammadAteeq/AQANB to support reproducibility and further research.

designed for evaluating models on lengthy narrative books, thereby enabling robust assessment of QA performance in extended and interconnected narrative contexts.

2) **An annotation-free framework for training the retriever:** We propose a novel, annotation-free framework that employs LLMs to generate challenging question-passage pairs focused on semantic content rather than token overlap, ensuring the retriever learns beyond surface-level token matching.
3) **A narrative-coherence scoring mechanism to enhance the retriever training:** We formulate retriever training as a regression problem, assigning scores inversely proportional to the distance from the most relevant passage, thereby capturing a broader narrative context and preserving story continuity.

The remainder of this paper is structured as follows. In Section II, we review the RAG approach, examine its challenges, and discuss recent progress in developing dense retrievers without annotations. Section III outlines our methodology, including details of the collected datasets and our proposed approach. In Section IV, we present and analyze our experimental findings. In Section V, we discuss the limitations. Finally, Section VI concludes the paper and suggests potential directions for future work.

II. LITERATURE REVIEW

Narrative QA entails deriving answers to questions by understanding the context, events, and relationships within narrative texts [11]. These texts, often exceeding 200K tokens,[2] present significant challenges for language models due to the computational demands of processing such extensive inputs. Current language models face difficulties stemming from the quadratic complexity of self-attention mechanisms [9] and the resource-intensive feedforward and projection layers applied to every token [7]. While recent advancements, such as LongT5 [12], CoLT5 [7], and FlashAttention [9], attempt to mitigate these limitations, they often rely on approximate attention mechanisms, which can compromise performance. Additionally, the architectures of these models remain computationally expensive when processing lengthy inputs [7]. A critical challenge identified by [8] is the underperformance of language models when key information is embedded in the middle of long inputs compared to when it appears at the beginning. This underscores the complexity of maintaining focus on relevant content throughout long contexts and highlights the need for strategies that reposition the relevant information to enhance performance in narrative QA tasks.

RAG [13] offers a promising approach to address the challenges associated with LLMs in handling lengthy inputs. RAG achieved competitive performance when employed in different applications [14], [15]. RAG leverages efficient retrievers to identify the most relevant text segments in response to a question. These selected passages are then processed by a language model with a limited context window, enabling the model to concentrate on the most pertinent information without processing the entire input. This approach significantly enhances computational efficiency. However, developing effective retrievers typically requires extensive datasets of question-passage pairs, which are often unavailable. This limitation hampers the ability of retrievers to accurately identify relevant information. Consequently, alternative methods for training dense retrievers without direct annotations are critical to advancing the capabilities of RAG-based systems.

LONGMEM [16] has emerged as a promising alternative for long-context modeling by equipping language models with an internal long-term memory mechanism. In contrast to RAG [13], which retrieves external passages using dense similarity search and typically relies on large annotated datasets of question-passage pairs, LONGMEM internally caches attention key-value pairs extracted from previous segments of the same input. During inference, a frozen backbone model is used to encode new inputs, while a lightweight, trainable SideNet performs memory retrieval and fusion through a memory-augmented attention layer. This decoupled design enables LONGMEM to efficiently preserve long-range coherence and retrieve from a significantly larger memory space (up to 65K tokens) without requiring retraining the backbone model. While this framework operates without needing explicitly annotated question-passage pairs during memory construction. However, fine-tuning still requires question-answer supervision to train the downstream QA models. Additionally, although LONGMEM can process up to 65K tokens of prior context, narrative books often far exceed this capacity, posing challenges in selecting and organizing the most relevant segments for effective long-term reasoning. Addressing these limitations is critical to fully leveraging LONGMEM for comprehensive narrative understanding.

Fusion-in-Decoder [17] is a generative open-domain question answering (ODQA) approach where each retrieved passage is independently encoded, and the decoder attends over all encoded passages to generate the final answer. While this method effectively aggregates information from multiple sources, applying it to NarrativeQA tasks poses significant challenges. Specifically, the approach heavily relies on the encoder to process each passage separately, which becomes computationally expensive, especially with large models exceeding one billion parameters. In contrast, RAG frameworks can use lightweight retrievers to pre-filter passages before passing a small set of selected tokens to a large reader model. For instance, our proposed retriever operates with just 177 million parameters, whereas the encoder component of models like mT5-large[3] exceeds 500 million parameters. Moreover, passing a large number

[2]For example, narrative books in the NarrativeQA dataset [11] and novels available at https://www.hindawi.org/books/categories/novels

[3]https://huggingface.co/google/mt5-large

of narrative segments to the encoder exacerbates the ''lost in the middle'' problem [8], where critical information situated mid-passage tends to be overlooked, diminishing answer quality.

Applying existing approaches to Narrative QA presents distinct challenges. Segmenting books and using Retrieval-Augmented Generation (Segmented-RAG) depends on retrievers trained with large question-passage datasets, which are often unavailable. Furthermore, it focuses only on local relevance between the question and segments, often ignoring the broader narrative coherence essential for understanding the narrative flow and the interconnected segments. Fusion-in-Decoder (FiD) requires independently encoding each passage, leading to high computational costs with large models and worsening the ''lost in the middle'' issue [8], where important mid-passage information is overlooked. LONGMEM [16] internally caches past context for retrieval but remains constrained by its memory capacity (up to 65K tokens), which is insufficient for full-length narratives, making it difficult to preserve global coherence across entire books. The subsequent sections will delve into recent methodologies proposed to train the retriever without direct supervision.

A. CROSS-LINGUAL LEARNING FOR DENSE RETRIEVAL

Cross-lingual transfer learning facilitates the adaptation of models trained in a monolingual context to operate effectively across diverse languages [18], [19]. This approach primarily aims to develop shared multilingual text representations by aligning linguistic features across languages [18], [20]. The process encompasses three key phases [20]: (1) constructing multilingual models, such as mT5 [21] and mBART [22], (2) fine-tuning these models for specific tasks using data in a source language, and (3) leveraging the fine-tuned models in zero-shot settings for a target language or further fine-tuning them on small-scale datasets in the target language. As highlighted in [23], pre-trained multilingual language models significantly reduce the dataset size required for effective performance by up to 80%. Moreover, [5] demonstrated the effectiveness of a two-stage fine-tuning approach for multilingual models. This method, which involves initial fine-tuning on an English narrative dataset followed by fine-tuning on a limited subset of Arabic translations (13% of the original dataset), achieves comparable performance to models trained on fully translated datasets. Consequently, this approach mitigates the high costs associated with full dataset translation or extensive language-specific annotations, underscoring its cost-efficiency and practical utility. UniBridge [24] enhanced the performance of cross -lingual transfer learning tasks for low-resource languages by introducing a method that systematically aligns the embeddings of target language tokens with those of the source language, leveraging both lexical overlap and semantic similarity. Similar studies have addressed resource constraints for other low-resource languages, such as Turkish [25], by repurposing existing datasets to simulate conversational contexts for Turkish, achieving substantial improvements in both retrieval and reading performance.

Cross-lingual transfer has also demonstrated substantial potential in multilingual dense retrieval, where retrievers trained on high-resource languages (e.g., English) generalize effectively to low-resource languages. For instance, [26] illustrated the benefits of fine-tuning multilingual transformers, such as mContriever [27], using machine-translated datasets. These strategies enable robust cross-lingual transferability, where relevance-matching capabilities learned in one language are seamlessly applied to others. Furthermore, the KEPT framework [28] advances cross-lingual dense retrieval by integrating structured knowledge from resources like Wikipedia and Wikidata, aligning semantic relationships across languages. KEPT employs contrastive learning during pre-training to map semantically similar queries and documents into a shared dense space, further enhancing its ability to generalize across linguistic boundaries.

B. SYNTHETIC DATA GENERATION WITH LLMs

LLMs have substantially advanced data augmentation strategies for training dense retrieval systems, effectively mitigating the common challenge of insufficient annotated data. Notably, InPars [29] and SPTAR [30] have been proposed to enhance retrieval effectiveness by leveraging LLMs for synthetic data generation. InPars employs manually crafted prompts to guide LLMs in generating synthetic queries from documents in an unsupervised manner. Although effective, its dependence on high-quality human-written prompts restricts adaptability across diverse domains. Conversely, SPTAR introduces a novel framework that utilizes soft prompt tuning—learnable embeddings trained on a small labeled dataset—to enable task-specific customization. However, this approach still necessitates a small number of annotated samples for each task.

In the realm of conversational QA, CONVERSER [31] employs in-context learning to synthesize passage-dialogue pairs with as few as six annotated examples, thereby generating conversational queries and passages. Similarly, CONVAUG [32] leverages LLMs to generate diverse conversational contexts through multi-level data augmentation strategies.

LLMs also enhance zero-shot dense retrieval approaches. HyDE (Hypothetical Document Embeddings) [33] is designed for zero-shot dense retrieval without reliance on relevance labels. HyDE decomposes retrieval into two key stages: first, it generates a hypothetical document via an LLM; second, it grounds this hypothetical document within the real corpus using a contrastive encoder. This two-step process capitalizes on generative capabilities to hypothesize relevant passages and employs contrastive learning to ensure document-document similarity in the embedding space.

SWIM-IR [34] expands LLM-driven data augmentation to low-resource languages by introducing synthetic datasets in multiple languages, offering a cost-effective and scalable approach for training multilingual dense retrieval models.

Despite these advancements, challenges remain in devising robust, domain-adaptive data augmentation strategies that maintain high retrieval performance across diverse tasks and languages.

C. UNSUPERVISED PRE-TRAINING AND FINE-TUNING

Recent approaches [35], [36] have shown that models can be trained effectively using pseudo-labels generated from unannotated data. The ART framework [35] generates pseudo-labels by leveraging a pre-trained language model. It uses questions and unpaired passages, treating retrieved passages as noisy approximations of the question. A pre-trained language model evaluates these passages to generate relevance scores, which act as pseudo-labels. The retriever is then trained to align its outputs with these pseudo-labels, improving its ability to retrieve relevant passages without requiring explicit question-passage annotations. Similarly, the UMR framework [36] also relies on pseudo-labels but focuses on multilingual tasks. It uses a generative language model to rank passages based on their relevance to questions and transfers this knowledge to train a dense retriever. This makes UMR particularly effective for handling low-resource languages where labeled data is scarce. Both methods showcase the power of pseudo-labeling in unsupervised training, enabling retrievers to perform well without the need for expensive labeled datasets.

D. UNIVERSAL TEXT EMBEDDINGS

Universal embeddings play a pivotal role in advancing retriever models, enabling them to generalize effectively across diverse tasks and domains. Unlike traditional task-specific approaches, these embeddings are crafted using heterogeneous, high-quality datasets that capture nuanced semantic relationships across varied contexts. The integration of synthetic data generated by LLMs further enhances the effectiveness of retrievers by providing pseudo-labeled data in multiple languages and domains, addressing data scarcity challenges. This rich combination of real-world and synthetic datasets allows retriever models to achieve superior performance in retrieval and passage-ranking tasks. The recent contributions for the universal text embedding appraoches can be categorized into three main categories: leveraging the large-scale and diverse datasets, advanced loss functions, and utilization of the LLM as a backbone of the retreiver.

1) LEVERAGING THE LARGE-SCALE AND DIVERSE DATASETS

The utilization of diverse, high-quality datasets is essential for developing effective retriever models. M3-Embedding [37] addresses critical challenges in multilingual and multi-functional text embeddings by supporting over 100 languages. It integrates various retrieval functionalities, improving data quality significantly during training. For general-purpose fine-tuning, the authors collected a large set of approximately 100 million unlabeled text pairs from multilingual datasets. They applied a filtering process to ensure the data was semantically meaningful and contextually rich. Additionally, the model utilized a labeled dataset of 838,465 text pairs for task-specific fine-tuning. This labeled data was derived from datasets like MIRACL [38], and synthetic data generated using GPT-based methods. AraDPR [1] is a retrieval model tailored specifically for Arabic text. Its development followed a carefully organized process to ensure the dataset's quality and relevance for Modern Standard Arabic applications. Articles were sourced randomly from Arabic Wikipedia to ensure topic diversity. Subsequently, Arabic-speaking annotators crafted questions by following clear guidelines, emphasizing factual accuracy and alignment with the provided content.

2) ENHANCEMENTS ON LOSS FUNCTION

The Angle-Optimized Embedding (AoE) [39] model improves semantic similarity tasks by addressing the challenges of cosine similarity, which struggles with vanishing gradients in saturation zones. AoE introduces a new method that splits embeddings into real and imaginary parts. The real part captures basic similarities, while the imaginary part focuses on small but important differences. AoE's key innovation is its angle-based loss function, which reduces angular differences between similar texts while keeping different texts distinct. Similarly, CoSENT [40] proposed a ranking-based loss to optimize cosine similarity rankings, improving sentence embedding alignment.

3) LLM BACKBONE UTILIZATION

In recent years, dense retrieval models have extensively utilized pre-trained language models such as BERT [41] and multilingual-T5 [21] as backbone encoders, effectively capturing semantic relationships between queries and passages. However, these models often face challenges in generalization, requiring substantial labeled data for fine-tuning to achieve strong performance in domain-specific tasks [10]. In contrast, larger models exhibit robust zero-shot generalization capabilities, delivering improved results without the need for domain-specific annotations [10]. For instance, Llama2Vec [42] leverages LLM embeddings for dense retrieval in an unsupervised manner, while BeLLM [43] incorporates bi-directional attention mechanisms to enhance semantic context comprehension. Despite these advancements, employing large LLMs as retrieval backbones presents notable challenges, including high computational costs, memory usage, and energy demands, which raise concerns about their feasibility and reliability for specialized retrieval tasks.

E. UTILIZING ARABIC LLMs IN READER PHASE

The second phase of RAG approach involves reading the retrieved passages to generate accurate answers. This section examines recent LLMs suitable for the Arabic language. AceGPT [44] introduced significant advancements in adapting LLMs to Arabic, leveraging diverse datasets alongside instruction tuning using Arabic questions and GPT4-generated responses. Furthermore, it incorporates

reinforcement learning with AI feedback (RLAIF) to align outputs with Arabic cultural values, ensuring nuanced understanding and contextually appropriate responses. AceGPT is built on the LLaMA2 backbone and fine-tuned on 7-billion and 13-billion parameter checkpoints. However, its context length is limited to 2,048 tokens, a constraint inherited from LLaMA2, which prevents it from processing entire books without the assistance of a retriever. Even with a retriever model, AceGPT faces limitations in handling the concatenated retrieved passages due to its input length constraints. Similarly, Dallah [45], a dialect-aware Multimodal Large Language Model (MLLM) designed specifically for Arabic, faces these limitations. Dallah is also built on the LLaMA2 backbone, featuring 7 billion parameters and a maximum context length of 2,048 tokens, which constrains its ability to fully leverage its capabilities for extended textual contexts.

The JASMINE models [46] represents a significant advancement in Arabic NLP, leveraging a Transformer-based GPT-Neo architecture, which replicates GPT-3, with a fixed context length of 2,048 tokens. The suite includes models with varying sizes—350M, 1.3B, 2.7B, and 6.7B parameters pre-trained on a diverse 235GB dataset that encompasses Modern Standard Arabic (MSA), Dialectal Arabic (DA), and Classical Arabic (CA). This extensive pretraining ensures linguistic diversity and equips the models with robust capabilities in few-shot and zero-shot learning.

The Gemma2 model, developed by Google, represents a cutting-edge advancement in large language model technology. It is designed to address a wide array of tasks, showcasing superior reasoning capabilities and efficient fine-tuning mechanisms. With its robust support for multilingual applications, Gemma2 excels in understanding context-rich queries, making it a valuable asset for both academic research and practical applications.[4] The SILMA[5] model, which utilizes the 9B checkpoint of Gemma2, has demonstrated exceptional performance, achieving the highest performance model on the Open Arabic LLM Leaderboard.[6] A key feature of the SILMA model is its ability to process inputs of up to 8,192 tokens, making it highly suitable for RAG applications. This capability enables the model to handle approximately 20 passages in a single input, assuming each passage consists of 400 tokens.

F. DATASETS FOR ARABIC QUESTION ANSWERING

Several datasets have been developed to facilitate Arabic open-domain QA tasks, with most efforts focusing on open-domain QA challenges. For instance, the ArabicaQA dataset [1] comprises 89,000 questions. This dataset, derived from Wikipedia through crowd-sourcing, is specifically designed to support open-domain QA in Arabic. Similarly, the Arabic-SQuAD dataset [3] includes 48,344 extractive question-answer pairs, generated via the translation of the English SQuAD dataset. Another notable dataset, ARCD [3], contains 1,395 extractive question-answer pairs tailored for Arabic QA tasks.

Beyond open-domain QA, a limited number of datasets have been developed to address QA tasks focused on religious texts, such as Hadith and Quranic scripture. Authors in [2] introduced HAQA and QUQA, two structured and annotated datasets that significantly enrich Arabic QA resources. HAQA, the first publicly available Hadith QA dataset, contains 1,598 question-answer pairs derived from canonical Hadith collections. Meanwhile, QUQA, the largest Quranic QA dataset to date, consists of 3,382 question-answer pairs covering approximately 47% of Quranic verses. Additionally, the QRCD dataset [47] offers 1,093 question-passage pairs focused on Quranic texts.

Despite these advancements, existing datasets predominantly focus on factoid questions, which limit their applicability to more complex QA tasks. Moreover, the average length of a Wikipedia article, approximately 690 tokens,[7] poses challenges for processing long-form content.

Developing QA datasets tailored to lengthy narrative books is crucial due to the unique challenges they present compared to datasets focused on factoid questions, such as those found in open-domain QA tasks. Factoid questions typically require concise answers derived from isolated, semantically relevant passages. In contrast, answering questions about narrative books demands a deeper understanding of the logical connections and interactions between multiple paragraphs. Often, passages that appear unrelated to the question on a semantic level may still need to be retrieved to construct a comprehensive sequence of events, providing the full context necessary for deriving accurate answers. This complexity highlights the importance of QA systems that can retrieve not only relevant passages but also those essential for maintaining narrative coherence. Such systems are indispensable for handling real-world scenarios involving lengthy and interconnected texts, where understanding the broader picture is key to answering challenging questions effectively.

The NarrativeQA dataset [6], widely recognized for its focus on lengthy narrative books, provides question-answer pairs specifically for English texts, making it a valuable resource for exploring QA in narrative contexts. However, to the best of our knowledge, no existing dataset addresses QA on narrative books in Arabic, except for the recently introduced Arabic NarrativeQA dataset [5]. However, this dataset is limited to books that do not exceed 2000 tokens.

III. METHODOLOGY

A. AQANB DATASET

We selected 103 publicly available Arabic narrative books from the Hindawi Foundation's online platform, which

[4]Official blog post from Google: https://blog.google/technology/developers/google-gemma-2/

[5]The model can be accessed at: https://huggingface.co/silma-ai/SILMA-9B-Instruct-v1.0

[6]For further details, see the Open Arabic LLM Leaderboard: https://huggingface.co/spaces/OALL/Open-Arabic-LLM-Leaderboard

[7]https://en.wikipedia.org/wiki/Wikipedia%3ASize_of_Wikipedia

provides a broad collection of literary works.[8]. We then divided the dataset into two subsets: 91 books were reserved for training and 12 books for evaluation. For training the retriever model, our proposed framework generated 7,736 questions, whereas 1,505 question–answer pairs were manually created by three native Arabic-speaking annotators for evaluation. To preserve the authenticity of the content, each book was obtained in its full-text form without modifications.

To ensure that AQANB contains books suitable for long-context reasoning, Arabic native speakers reviewed the metadata and summaries of all candidate books before selection. This process was used to confirm that each book had sufficient narrative length, depth, and interconnected plot developments that would enable the creation of challenging questions requiring reasoning across multiple events. Books that merely described facts, presented simple or linear narration, or lacked enough narrative complexity were excluded. By applying these explicit selection criteria, the final set of books in AQANB provides the rich and coherent narrative structure needed for evaluating long-context comprehension and reasoning.

Annotators were encouraged—though not strictly required—to create questions that went beyond simple factual recall, emphasizing inference, summarization, or multi-hop reasoning. Corresponding answers were provided in Arabic. To enable deeper analysis, each question in the dataset was assigned an *intent category* reflecting its primary goal or focus. This classification scheme helps evaluate how well QA systems address different dimensions of narrative understanding and is organized into seven distinct categories.

1) **Setting**: Questions about the time, place, and general context where the story events occur.
2) **Character**: Questions that focus on the people in the story, including who they are, their roles, and their relationships with others.
3) **Prediction**: Questions that use clues from the story to guess what might happen next or how a character might behave.
4) **Causal Relationship**: Questions that explore why something happened in the story and how one event led to another.
5) **Outcome Resolution**: Questions about the resolution or outcomes of conflicts, troubles, or challenges faced by the story's characters.
6) **Feeling**: Questions that look at the emotions, reactions, or experiences of the characters based on what happens in the story.
7) **Action**: Questions about specific things the characters do, including their decisions and the events they cause.

[8]Hindawi.org is a non-profit organization dedicated to promoting knowledge and culture among Arabic speakers. According to its privacy policy, the platform provides free access to its content, making it an invaluable resource for educational and learning purposes. https://www.hindawi.org/books/categories/novels

TABLE 1. Statistics of the AQANB dataset: token information.

Category	Min	Mean	Max	Std Dev
Tokens per Book				
Training Books	18,216	112,222	434,815	68,467
Evaluation Books	72,463	149,227	215,712	37,930
Tokens in QA Pairs				
Question Length	7	17.74	44	4.20
Answer Length	3	15.26	52	4.82

TABLE 2. Statistics of the AQANB Dataset: question categories and types.

Category	Count (%)	Description
Question Categories		
Action	365 (24.2%)	Behaviors, decisions
Causal Relationship	437 (29.0%)	Cause-and-effect
Feeling	170 (11.3%)	Emotional states
Character	207 (13.7%)	Roles, traits
Outcome Resolution	136 (9.0%)	Conflict resolution
Prediction	80 (5.3%)	Future actions
Setting	110 (7.3%)	Time, place, context
Answer Type		
Explicit	737 (49.0%)	Direct extraction
Implicit	768 (51.0%)	Paraphrasing is required
Reasoning Type		
Single-Hop	1,033 (68.6%)	Localized context
Multi-Hop	472 (31.4%)	Multiple paragraphs
Coreference		
Without Coreference	987 (65.6%)	Does not require coreference
With Coreference	518 (34.4%)	Requires coreference resolution

Beyond the *intent* categories, each question was also labeled based on whether the answer is *explicitly* or *implicitly* retrievable from the text:

- **Explicit Questions**: These questions can be answered directly using tokens present in the text. Answering them typically involves locating and extracting relevant information.
- **Implicit Questions**: These questions cannot be answered through direct token extraction from the book. Instead, they require generating new tokens or paraphrasing the original content.

Categorizing questions as explicit or implicit is crucial for understanding the complexity of the QA task. In general, implicit questions require a model to perform more advanced reasoning and summarization.

Questions were further classified based on whether they necessitated *multi-hop reasoning*:

- **Single-hop questions**: These are questions whose answers can be found in a single paragraph.
- **Multi-hop questions**: These questions require integrating details scattered across multiple paragraphs to derive an answer.

Some questions require coreference resolution, which involves identifying references to the same entity across different parts of the text. Coreference tests a model's ability to track entities and maintain narrative coherence such as characters or events.

For further details, we followed prior works such as [5] and [48] to categorize the questions based on their intent and implicitness. In addition, we introduced two new dimensions

for categorization: multi-hop versus single-hop questions, and the presence of coreference resolution.

Tables 1 and 2 provide detailed statistics of the AQANB dataset. Table 1 highlights that the average token count for books in the evaluation dataset is 149,227, demonstrating its suitability for assessing QA models on lengthy books. This is particularly relevant for tasks that require understanding interactions across multiple paragraphs and capturing the broader context essential for generating accurate and comprehensive answers. Notably, some books exceed 150,000 tokens, imposing limitations on the applicability of most existing models. For instance, GPT4 has a maximum token limit of 128,000 .[9] Table 2 illustrates the comprehensive diversity of question categories within the dataset, encompassing a wide spectrum of thematic areas. Notably, the dataset achieves a balanced distribution between explicit and implicit answer types, ensuring robust evaluation of both direct information extraction and paraphrasing capabilities. The reasoning complexity is systematically assessed through a combination of single-hop and multi-hop questions, enabling thorough evaluation of both localized and cross-paragraph reasoning. Furthermore, a significant proportion of the dataset (34.4%) incorporates coreference resolution, highlighting its utility in evaluating entity and event tracking across the text.

Table 3 showcases a selected excerpt from a broader narrative, illustrating how logical relationships between paragraphs contribute to understanding the story and answering questions. Although the complete narrative is extensive, the table highlights pivotal interactions between the characters Wang Lung and his uncle that are relevant to provided question. The sequence of events across the three paragraphs captures the cause-and-effect relationship and the escalating emotional tension. In this example, the three interconnected paragraphs collectively explain Wang Lung's anger toward his uncle; however, no single paragraph provides a complete answer if presented alone to the reader along with the question. Paragraph I conveys Wang Lung's frustration through implicit cues, such as the action of ''dropping his axe and suddenly shouting'' which requires a robust retriever to infer anger without explicit mention of the emotion. Paragraph II introduces dialogue (''What do you want me to do?''), providing additional context about the uncle's threats, which heighten the tension. Paragraph III ultimately uncovers the core reason behind Wang Lung's anger—an inappropriate financial demand from his uncle. This example underscores the complexity of synthesizing dispersed information and inferring underlying motives across the narrative. To generate an accurate, context-aware answer, a successful QA system must integrate implicit emotional signals, escalating threats, and financial demands, effectively reasoning across the interconnected paragraphs.

[9]Official documentation of OpenAI: https://help.openai.com/en/articles/7102672-how-can-i-access-gpt-4-gpt-4-turbo-gpt-4o-and-gpt-4o-mini

While this example demonstrates the narrative complexity within contiguous paragraphs, the dataset also contains more challenging scenarios where the related paragraphs are not contiguous, requiring the retrieval system to bridge gaps and identify connections across dispersed sections of the text.

B. SYNTHETIC NARRATIVE-AWARE RETRIEVER

Synthetic Narrative-Aware Retriever (SNAR) is an innovative framework for training the retriever, designed to overcome the scarcity of costly manual annotations. SNAR leverages an LLM to generate event-centric question-passage pairs, enabling the retriever to capture the semantic relevance of the question rather than relying on superficial token overlap. The overall workflow of SNAR is summarized in Algorithm 1 and illustrated in Figure 1, which together outline its five main stages: segmentation of the narrative text, event extraction, event-based question generation, pseudo-labeling, and retriever training.

Algorithm 1 The Main Steps for Building SNAR Retriever

Input: Narrative book $\mathcal{B}$
Output: Trained retriever model
Step 1: Segmentation
Segment $\mathcal{B}$ into non-overlapping passages $\mathcal{P} = \{p_1, p_2, \ldots, p_K\}$ of 400 tokens each
Step 2: Events Extraction
for each passage $p_k \in \mathcal{P}$ **do**
    Use the LLM to extract salient events e_k from p_k // Each e_k represents the key narrative developments in p_k
end for
Step 3: Event-Based Question Generation
for each e_k **do**
    Prompt the LLM to generate a question q_k given e_k
end for
Step 4: Pseudo Labeling
for each question q_j and passage p_i **do**
    Assign relevance score $s(q_j, p_i)$ using Eq. 3
end for
Obtain pseudo-label matrix $S = [s(q_j, p_i)]$
Step 5: Retriever Training
Select top H passages with highest s scores
Encode questions and passages using either:
- Bi-Encoder: independently encode q_j and p_i to compute similarity via cosine function
- Cross-Encoder: jointly encode concatenated (q_j, p_i) pair for fine-grained interaction

Compute predicted similarity $\hat{s}_i(q_j, p_i)$ for each pair
Minimize Mean Squared Error loss as defined in Eq. 4:
Save trained retriever model

1) SEGMENTATION

The input narrative, such as a book, is segmented into non-overlapping passages $(p_1, p_2, \ldots, p_k)$ of 400 tokens. The segmentation is performed without enforcing sentence or

TABLE 3. Example a question-answer pair, and the context. It demonstrates the narrative complexity within contiguous paragraphs.

Context	Question	Answer
أسقط وانج لنج فأسه وصاح فجأة، وهو يحملق في عمه: إن كنت أملك الآن حفنة من الفضة، فذلك لأني أشتغل أنا وزوجتي، ولا نجلس بدون عمل كغرينا، أو نقضي الوقت في القيل والقال، تاركني حقولنا تتحول إلى أعشاب ضارة، أو نترك أطفالنا نصف جياع! فصاح عمه في صوت ثائر: سأخبر جميع القرية بما قلتَه. سأخبر به القرية، سأتحدث به في القرية ... وأخذ يكرر هذه العبارة مرات ومرات، حتى قال وانج لنج مكرها: وماذا تريدني أن أفعل؟ تغير مسلك عمه في الحال، وابتسم، ووضع يده على ذراع وانج لنج، قائلا في دعة تغرير بضع قطع فضية في هذه الكف الفقرية؛ عشر قطع، أو حتى تسع مثلا، فأستطيع البدء بعمل الترتيب اللازم مع إحدى الخاطبات بشأن عبدتي. Pragraph1: Wang Lung dropped his axe and suddenly shouted, staring at his uncle: "If I now have a handful of silver, it is because my wife and I work hard, and we don't sit idly like others, wasting time gossiping, leaving our fields to turn into weeds, or letting our children go half-starved! Pragraph2: His uncle shouted in an angry voice: "I will tell the whole village what you said. I will tell the village, and I will speak about it in the village..." He kept repeating this phrase over and over until Wang Lung reluctantly said: "What do you want me to do?" Pragraph3:His uncle's tone changed immediately. He smiled, placed his hand on Wang Lung's arm, and said gently: "A few silver pieces in this poor hand; ten pieces, or even nine, perhaps, and I can arrange matters with one of the matchmakers regarding my maid."	ما هو السبب وراء غضب وانج لنج من عمه؟ What is the reason behind Wang Lung's anger towards his uncle?	لأن عمه طلب منه المال بطريقة غير لائقة. Because his uncle asked him for money in an inappropriate manner.

paragraph boundaries. The choice of 400 tokens per passage ensures compatibility with the input constraints of BERT-based models. Additionally, longer passages pose significant challenges in embedding computation, as aggregation methods such as mean pooling often fail to adequately capture subtle semantic distinctions, leading to potential loss of contextual nuances [49].

2) EVENTS EXTRACTOR

For a given passage p_k, the LLM identifies and extracts salient events e_k, which encapsulate the core narrative and essential semantics of the text. Specifically, e_k is a concatenated string of salient events produced when p_k is input into the LLM and prompted accordingly.

To extract these events, we use the following prompt: *"Please read the narrative and extract the main events. Summarize them in a single coherent paragraph using concise sentences, such that the summary represents approximately 20% of the original text. The output should be written in Arabic."*.This prompt guides the model to generate a connected summary of key events. The 20% constraint is used to force the model to be concise and to prioritize only the most important developments in the passage. This helps ensure that the generated summary captures multiple key events rather than overly focusing on minor or isolated ones. Consequently, this step ensures that the subsequent generation of questions aligns closely with the intrinsic meaning and context of the passage. Figure 2 illustrates an example of salient events (e_k) extracted using the GPT-4o model .[10]

[10]gpt-4o-2024-08-06 version. https://openai.com/index/gpt-4/

3) QUESTIONS GENERATOR

These extracted events serve as prompts for the LLM (GPT-4o in our case) to generate questions designed to probe the main events. This design avoids reusing exact tokens from the passage to construct the question. Thus, the questions emphasize higher-level comprehension of the passage rather than surface-level word matching. As a result, the model must attend to the meaning of the passage rather than rely on lexical overlap.

To guide the LLM in generating meaningful and reasoning-based questions, we use the following prompt: *"Read the narrative text below and create a concise, challenging question that requires strong reasoning skills and reflects your understanding of its key points"* This prompt encourages the model to generate questions that test comprehension of complex ideas and connections rather than superficial facts. The following is an example of a generated question utilzing the extracted events in Figure 2.

What is the fundamental contradiction in Cotherstone's behavior toward Kaitly, and how does this contradiction reveal his psychological state?

ما التناقض الأساسي في تصرفات كوذرستون تجاه كايتلي، وكيف يكشف هذا التناقض عن حالته النفسية؟

This event-based question generation approach offers several key advantages for enhancing contextual relevance and semantic depth in retrieval. First, it advances contextual alignment by requiring the model to retrieve passages that reflect not only individual events but also the interplay of multiple narrative states and developments. Consequently, the retriever is challenged to comprehend the broader

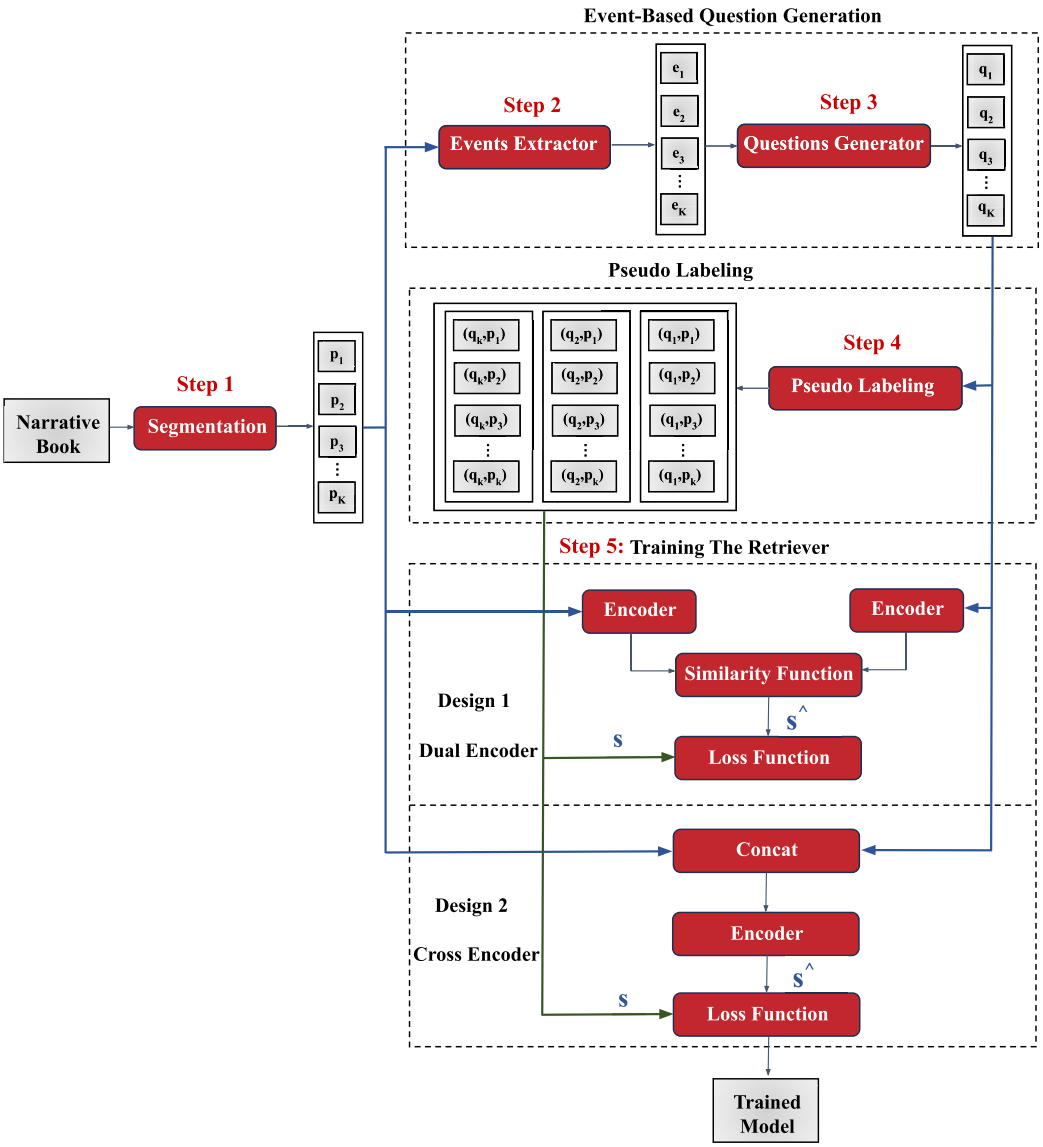


FIGURE 1. SNAR framework, illustrating segmentation, event extraction, event-based question generation, pseudo-labeling, and retriever training via bi-encoder and cross-encoder architectures.

narrative context in which themes such as fear, conflict, and concealment are situated. Second, by explicitly avoiding the reuse of tokens from the original passage, the retriever is trained to prioritize deeper semantic associations between the question and the passage, rather than relying on superficial lexical overlaps. Figure 2 presents another generated question that encapsulates Cotherstone's internal conflict and his fear of Kightley—both central themes of the passage. In this example, tokens highlighted in red are derived from the extracted events used to generate the question but do not appear in the passage itself. This deliberate vocabulary separation ensures that the retriever must rely on semantic comprehension rather than direct token matching, ultimately promoting more meaningful and contextually aware retrieval.

4) PSEUDO LABELING

We assign pseudo-label scores to passages in the following manner:

1) We formally define the *central passage* (p_{center}) as the passage explicitly utilized by the LLM during the process of generating the corresponding question. We assume this passage holds the highest relevance to the generated question. Consequently, it is assigned a full relevance score of 1, as defined in Equation 3. As illustrated in Figure 1, the *central passage* (p_{center}) corresponds to the condition where the index of the question aligns with the index of the passage. Let j represent the index of the question and i represent the

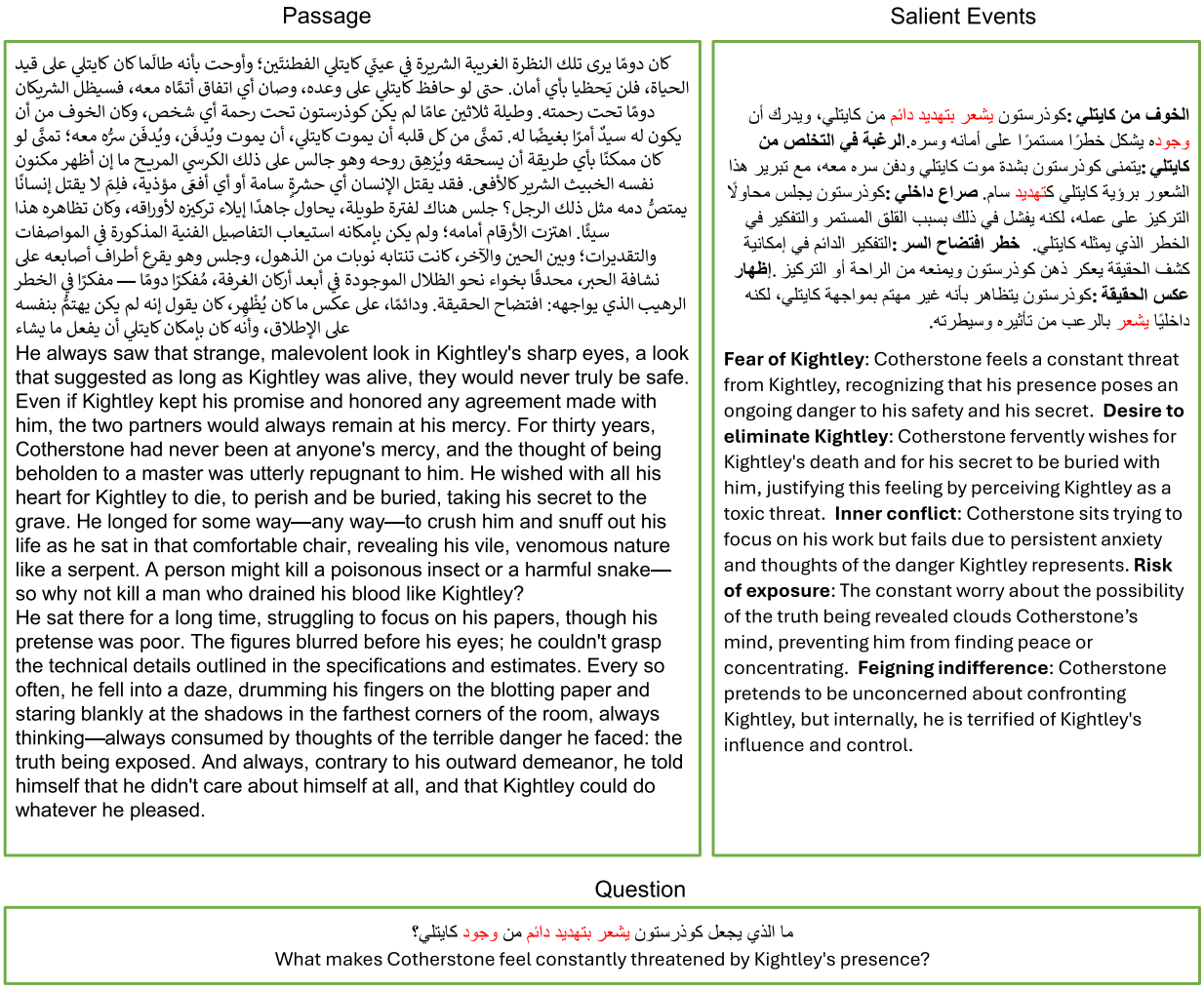


FIGURE 2. Illustration of the event-based question generation approach. Starting from finding the key narrative elements from the original passage. Then, these extracted events serve as prompts for generating questions that assess deeper comprehension beyond lexical overlap. In this example, highlighted tokens in red appear in the question but are absent from the passage, compelling the model to rely on semantic understanding rather than direct token matching.

index of the passage. The pair (q_j, p_i), when $j = i$, indicates that the question q_j was generated using the salient events of passage p_i. Conversely, when $j \neq i$, the question is not directly generated from the passage. However, we hypothesize that such passages should not be treated as strictly negative examples. Instead, we consider them as candidate passages with lower relevance, assigning them a reduced relevance score to reflect their potential contextual contribution.

2) All other passages (when $j \neq i$) are assigned scores based on two factors:

- Narrative coherence score: The proximity of each passage to the central passage which is quantified using an exponential distance weight, defined by 1:

$$w_i = \exp\Big(-\alpha \cdot \text{dist}(p_i, p_{\text{center}})\Big), \tag{1}$$

where $\text{dist}(p_i, p_{\text{center}})$ represents the absolute distance between the indices of passage p_i and the central passage p_{center}. The parameter α is a hyperparameter that governs the influence of adjacent passages. Smaller values of α result in higher contributions from passages closer to p_{center}, thereby enhancing the ability to reconstruct a comprehensive narrative by leveraging the context provided by nearby passages.

- The semantic relevance of each passage to the question which is quantified using cosine similarity 2:

$$c(q_j, p_i) = \cos(\mathbf{Q}_j, \mathbf{P}_i) \in [0, 1], \tag{2}$$

where $\mathbf{Q}_j$ is the embedding vector of the question q_j, and $\mathbf{P}_i$ is the embedding vector of the passage p_i.

The embeddings are computed using the OpenAI small embedding model.[11]

To balance between the distance (narrative coherence) and semantic relevance, we employ the following scoring function 3:

$$s(q_j, p_i) = \begin{cases} 1, & \text{if } j = i \\ \beta \cdot w_i + (1-\beta) \cdot c(q_j, p_i), & \text{otherwise} \end{cases} \quad (3)$$

where $\beta \in [0, 1]$. This scoring function ensures the following:

1) **Coherence of retrieved passages.** Adjacent passages to the central one may offer critical context or narrative continuity, even if they do not match the question's semantics perfectly.
2) **High relevance for direct answers.** Passages that are semantically closer to the question get higher scores when β is lower, reinforcing their importance for answering the question directly.
3) **Robust pseudo-labeling.** The model learns to consider both key (highly relevant) passages and their neighboring passages in a single narrative, producing better retrieval and downstream reasoning.

This strategy pseudo-labels passages to effectively capture broader narrative events and contextual interactions while ensuring relevance to the question. Unlike approaches that expand individual passage lengths to include more context, our method dynamically balances the need to construct a coherent narrative sequence with the importance of retrieving the most relevant parts of the text. Increasing passage lengths presents significant drawbacks. First, computing embeddings for longer passages often leads to challenges, as aggregation methods like mean pooling struggle to capture nuanced semantic distinctions [49], degrading the precision of similarity computations. Second, transformer-based encoders face strict token limits, and longer passages risk truncation, causing a loss of critical information. Finally, reader models, constrained by fixed context windows, face challenges in incorporating a diverse range of retrieved information when passages are excessively long. Extending passages to a fixed length not only reduces the number of passages the reader can process simultaneously but also increases the likelihood of including non-relevant information. This diminishes the effectiveness of the reader by diverting attention from passages that are most pertinent to the question, thereby impeding its capability to extract precise and accurate answers.

5) TRAINING THE RETRIEVER

We employ two retrieval methodologies: the bi-encoder and cross-encoder approaches. The bi-encoder framework follows the dense retrieval paradigm, where a single Transformer model is fine-tuned to encode both questions and passages into dense vector representations while sharing parameters across both encoders [50]. Fixed-dimensional vectors are generated through mean pooling over the token embeddings, and these representations are compared using cosine similarity. This design is highly scalable for large collections, as passage embeddings can be precomputed and cached, enabling efficient reuse across multiple questions during inference. Conversely, the cross-encoder processes question-passage pairs jointly by concatenating their tokens, enabling the model to capture more nuanced interactions between the question and the passage. While this often results in more precise relevance scores, it incurs significantly higher computational costs due to the need for a separate forward pass for each candidate passage. In our experiments, we systematically compare these approaches in terms of retrieval performance and computational efficiency, providing insights into their trade-offs.

The retriever training is formulated as a regression problem in which a continuous score is assigned to each passage according to Eq. (3). The model is optimized using the Mean Squared Error (MSE) loss L_{MSE}, defined in Eq. (4):

$$L_{\text{MSE}} = \frac{1}{J} \sum_{j=1}^{J} \left(\frac{1}{H} \sum_{i=1}^{H} \left(s(q_j, p_i) - \hat{s}(q_j, p_i) \right)^2 \right), \quad (4)$$

where J denotes the number of questions in the batch, H is the number of selected passages per question, $s(q_j, p_i)$ represents the score computed from Eq. (3) for the i-th passage of question q_j, and $\hat{s}(q_j, p_i)$ is the predicted score generated by the model (see Figure 1). To reduce the computational cost of training, we select the top H passages with the highest s scores, thereby avoiding the need to encode all passages using cross-encoder or dual-encoder models. This strategy enables the model to focus on distinguishing among the most challenging passages that are semantically closest to each question.

IV. RESULTS AND DISCUSSION

Section IV-A defines the evaluation metrics used to assess the performance of the proposed retrievers within the RAG system. Section IV-B introduces state-of-the-art retrievers that serve as baselines for comparison with our approach. In Section IV-C, we present a detailed performance comparison, demonstrating the superiority of our retrievers, which stems from their event-focused question generation and coherence-based scoring for pseudo-labeling—effectively addressing the second and third research objectives.

Importantly, the manuscript incorporates three ablation studies to validate the contribution of each core component. Section IV-D (Ablation Study I) evaluates the effectiveness of the annotation-free retriever training framework by removing the narrative coherence component, directly addressing the second research objective. Section IV-E (Ablation Study II) isolates the impact of narrative coherence scoring by varying its weighting parameters, thereby supporting the third research objective. Finally, Section IV-F (Ablation Study III) examines the proposed narrative-aware training methodology by comparing conventional binary labeling with the SNAR Pseudo Labeling strategy.

[11] https://openai.com/index/new-embedding-models-and-api-updates/

Section IV-G compares our retriever against methods that retrieve lengthy passages. Our approach prioritizes the retrieval of contiguous, contextually relevant passages, thereby preserving narrative coherence without necessitating the retrieval of excessively long passages—further contributing to the third research objective. Section IV-H analyzes the computational efficiency of SNAR-GPT4o, demonstrating its ability to significantly reduce computational costs while maintaining high retrieval performance, thus addressing both the second and third research objectives. Section IV-I investigates the sensitivity of the retriever to various hyperparameters. Finally, Section IV-J provides a comprehensive performance analysis across various question types, identifying key challenges in handling implicit, multi-hop, and coreference-based questions. This analysis underscores the critical role of the AQANB dataset in evaluating the advanced reasoning capabilities of language models, directly addressing the first research objective.

A. EVALUATION METRICS

Given that the evaluation dataset does not provide ground-truth passages for each question, the performance of all retrievers is assessed by integrating them into a RAG system. A unified reader is employed across all retrievers to ensure consistency in answer generation. The quality of the generated answers is then evaluated using established metrics. In alignment with prior studies [6], [7], [11], [12], [51], the following evaluation metrics are employed to assess the quality of the generated answers:

- **ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation - Longest Common Subsequence)** [52]: ROUGE-L is widely recognized in summarization and generative QA tasks for measuring the similarity between generated and reference texts. It focuses on the Longest Common Subsequence (LCS), identifying the sequence of words appearing in both texts in the same order, although not required to be contiguous. By capturing the structural and semantic coherence of texts, ROUGE-L computes an F-score based on both precision (the proportion of matched words in the generated text) and recall (the proportion of matched words in the reference text). This ensures a comprehensive assessment of the alignment between the generated output and the reference.
- **BLEU (Bilingual Evaluation Understudy)** [53]: BLEU serves as a tool for evaluating generative QA outputs through the measurement of n-gram precision. This precision measures the proportion of n-grams present in the generated text that correspond to those in the reference text. The metric evaluates the capability of the generated content to replicate the reference material, ensuring the preservation of structural integrity. Multiple variants of this metric exist, depending on the n-gram employed; for instance, BLEU-1 utilizes 1-grams, whereas BLEU-4 employs 4-grams.
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)** [54]: METEOR offers a more nuanced evaluation by incorporating synonym matches alongside exact word matches, aligning generated and reference words while penalizing order misalignments. This metric combines precision and recall into a single F-score, making it particularly suitable for generative QA tasks where semantic similarity is critical.
- **F1 Score**: This metric evaluates unigram overlaps between the generated and reference answers.

B. BASELINES

To evaluate the performance of our proposed models, we compare them against several baseline retrievers: **mContriever**, **M3**, **AraDPR**, **openai-small**, and **openai-large**. All these baselines utilize a bi-encoder architecture, where a shared encoder generates dense vector representations for both questions and passages. The primary distinction among these retrievers lies in their embedding strategies and training methodologies.

AraDPR [1] is a retriever specifically tailored for Arabic text retrieval. It is trained on the Arabic Wikipedia corpus and leverages pre-trained Arabic transformers such as AraBERT. AraDPR employs contrastive learning techniques to address the morphological and syntactic complexities inherent in the Arabic language. This model has demonstrated strong performance in open-domain retrieval and machine reading comprehension (MRC) tasks, effectively capturing the nuances of Arabic text.

GATE [55] is a recently introduced Arabic text embedding model that achieves state-of-the-art performance in many semantic similarity tasks. It employs advanced training strategies and a hybrid loss informed by Arabic triplet data, enabling it to capture subtle semantic nuances more effectively.

mContriever, the multilingual variant of Contriever [27], focuses on unsupervised dense retrieval. It employs augmentation techniques such as random cropping to create positive and negative pairs for training. This approach enables mContriever to excel in zero-shot and few-shot retrieval scenarios, making it a competitive choice for tasks with limited labeled data. Its emphasis on multilingual support further enhances its utility in diverse linguistic contexts.

M3 [37] functions as a multilingual retriever characterized by Multi-Linguality, Multi-Functionality, and Multi-Granularity, specifically developed to facilitate dense, sparse, and multi-vector retrieval methods. It utilizes a self-knowledge distillation framework that synthesizes scores derived from disparate retrieval paradigms, thereby augmenting its adaptability across different languages and tasks. This holistic methodology guarantees that M3 consistently achieves superior performance across a variety of retrieval contexts.

Jina-V3 [56] is one of the state-of-the-art multilingual embedding models, notable for its ability to support long-context retrieval and a broad range of semantic tasks.

With 570 million parameters, it accommodates context lengths of up to 8,192 tokens, enabling reliable encoding of extended documents and complex discourse structures. The model incorporates task-specific Low-Rank Adaptation (LoRA) modules that specialize its representations for various downstream applications, including query–document retrieval, clustering, classification, and text matching.

OpenAI's text-embedding models, including **embedding-3-small** and **embedding-3-large**, are state-of-the-art tools for transforming text into high-dimensional numerical vectors. These embeddings facilitate a range of tasks such as semantic search, clustering, and classification. The small model is optimized for efficiency, producing embeddings with 1,536 dimensions, while the large model is designed for superior performance, generating embeddings with up to 3,072 dimensions. These models capture intricate textual nuances and achieve competitive results on benchmarks such as MIRACL [38]. Although the specific architectural details of these models remain proprietary, they are part of the GPT series and are fine-tuned for embedding tasks, leveraging transformer-based architectures to deliver high-quality results.

C. COMPARISON WITH STATE-OF-THE-ART

Table 4 presents a detailed performance comparison of the proposed SNAR retrievers (SNAR-BI and SNAR-CROSS) against several state-of-the-art retrieval methods. Given that the evaluation dataset does not provide ground-truth passages for each question, the performance of all retrievers is measured using a unified reader model, SILMA .[12] This approach ensures that any differences in the reported evaluation metrics are exclusively due to the retrieval capabilities of each model, eliminating confounding factors from the reader's variability. SILMA was specifically chosen for this evaluation because it achieved the best results on the Open Arabic LLM Leaderboard ,[13] highlighting its suitability for comprehending the question and the retrieved passages. Furthermore, the number of retrieved passages and the passage length are held constant across all methods. The results are organized into three categories: OpenAI models, open-source models, and the proposed models, where SNAR-BI and SNAR-CROSS incorporate bi-encoder and cross-encoder architectures, respectively.

All retrievers presented in Table 4, except SNAR-CROSS, employ a bi-encoder architecture in which the same encoder is utilized for both the question and passage representations. These models rely on cosine similarity as the scoring function to compute the relevance between passage and question embeddings. The retrieval performance is therefore directly tied to the encoder's ability to generate high-quality embeddings that effectively capture the semantic content of both the passage and the question.

[12]https://huggingface.co/silma-ai/SILMA-9B-Instruct-v1.0

[13]https://huggingface.co/spaces/OALL/Open-Arabic-LLM-Leaderboard

TABLE 4. Performance comparison of the proposed SNAR retrievers against various state-of-the-art retrieval methods. All results are generated using the same reader model, SILMA. Additionally, the number of retrieved passages is set to 5 in all retrievers. This ensures that any variation in the reported evaluation metrics (F1, BLEU-1, ROUGE-L, METEOR) is solely attributable to the retrieval effectiveness of each model. The proposed retrievers, SNAR-BI and SNAR-CROSS, leverage bi-encoder and cross-encoder architectures, respectively.

Retriever	F1	BLEU-1	ROUGE-L	METEOR
OpenAI Models				
openai-small	18	13.6	17.5	13.7
openai-large	19.6	14.8	19.1	15.1
Open-Source Models				
mcontriever [56]	16.9	12.7	16.4	12.6
M3 [37]	18.5	13.7	18.1	13.7
Jina-V3 [57]	19.4	14.6	18.9	14.8
AraDPR [1]	14.1	10.6	13.7	9.8
GATE [55]	16.9	12.6	16.3	12.5
Proposed Models				
SNAR-BI	20.5	15.4	20.0	15.9
SNAR-CROSS	21.9	16.7	21.4	17.6

Our proposed SNAR-BI and SNAR-CROSS retrievers exhibit significant improvements over the baseline retrievers, with substantial performance gains across all evaluation metrics. When compared to the strongest OpenAI baseline, openai-large, both SNAR variants demonstrate superior results, achieving higher ROUGE-L scores (+0.9 for SNAR-BI and +2.3 for SNAR-CROSS) and METEOR scores (+0.8 for SNAR-BI and +2.5 for SNAR-CROSS). These improvements can be attributed to the effectiveness of our training strategy, which incorporates an event-focused question generation and proposed pseudo-labeling mechanism. Notably, OpenAI embeddings are computed via an API for the question and each candidate passage separately, with relevance determined using cosine similarity. This inherently limits flexibility, as it does not allow for a cross-encoder design where question-passage pairs can be concatenated and jointly processed to produce a relevance score.

When compared to open-source models, SNAR-BI shares the same design and achieves consistent gains over the strongest open-source baseline, Jina-V3, with improvements of +1.1 in ROUGE-L and +1.1 in METEOR. This highlights the robustness of SNAR-BI's retrieval capabilities, even when the bi-encoder architecture paradigm is utilized by both. SNAR-CROSS further amplifies these improvements, achieving the highest scores across all metrics due to its cross-encoder architecture, which enables it to capture richer interactions between questions and passages. These results collectively underscore the effectiveness of the event-focused question generation and pseudo-labeling strategy in both architectures.

The performance of AraDPR [1], a retriever specifically trained on Arabic texts, demonstrates the challenges of applying models trained on Wikipedia-like corpora to narrative-based tasks. AraDPR has shown strong results in open-domain retrieval and machine reading comprehension (MRC) tasks. However, its relatively lower performance in our evaluation highlights the inherent challenges of narrative texts compared to other content like Wikipedia. Narrative

books often contain passages that are semantically similar, making it difficult to discriminate between them. This underscores the importance of designing retrievers that can effectively handle deeper semantic relationships.

Table 4 shows that SNAR-CROSS demonstrates superior retrieval performance (F1: 21.9, BLEU-1: 16.7, ROUGE-L: 21.4, METEOR: 17.6) compared to SNAR-BI (F1: 20.5, BLEU-1: 15.4, ROUGE-L: 20.0, METEOR: 15.9). This improvement in SNAR-CROSS is attributed to its cross-encoder design, which jointly processes question-passage pairs to capture fine-grained interactions, resulting in more precise relevance scores. However, when both retrievers are run on the same server equipped with an NVIDIA RTX 6000 Ada GPU, this approach comes at a significant computational cost, with an execution time of 0.64 seconds per question, as it requires a separate forward pass for each candidate passage. In contrast, SNAR-BI achieves remarkable efficiency with an execution time of only 0.03 seconds per question. This efficiency stems from its bi-encoder design, where the embeddings of passages are precomputed and cached. These cached embeddings can be reused across multiple questions, significantly reducing the computational overhead during inference.

D. ABLATION STUDY 1: EFFECTIVENESS OF THE PROPOSED ANNOTATION-FREE FRAMEWORK ON RETRIEVER TRAINING

To evaluate the effectiveness of the proposed annotation-free approach, which aligns with the second research objective, we exclude narrative coherence from the pseudo-labeling process by setting $\beta = 0$ in Equation 3. Table 5 presents a comparative analysis of a bi-encoder retriever trained using our approach against other bi-encoder retrievers. Notably, our method outperforms the strongest open-source baseline, M3, with improvements of +0.9 in ROUGE-L and +1.4 in METEOR. These results demonstrate the robustness of the proposed approach in training retrievers without reliance on human-collected annotations.

E. ABLATION STUDY II: EFFECTIVENESS OF THE PROPOSED NARRATIVE COHERENCE SCORING ON RETRIEVER PERFORMANCE

The results in Table 6 demonstrate the effectiveness of incorporating narrative coherence scoring in retriever training.

TABLE 5. Performance evaluation of the proposed annotation-free retriever training approach, excluding narrative coherence scoring in pseudo-labeling. To ensure a fair and consistent evaluation, all retrievers retrieve five passages, and the same reader model, SILMA, is employed to comprehend the retrieved passages and generate answers. The table reports key performance metrics, including F1, BLEU-1, ROUGE-L (R-L), and METEOR scores. Bolded values indicate the highest scores achieved for each respective metric.

Retriever Method	F1	BLEU-1	R-L	METEOR
mCONTRIEVER-BI [56]	16.9	12.7	16.4	12.6
M3-BI [37]	18.5	13.7	18.1	13.7
SNAR-BI(w/o Coh.)	**19.6**	**14.6**	**19.0**	**15.1**

TABLE 6. Effectiveness of the proposed narrative coherence scoring on retriever training. The table evaluates the impact of incorporating narrative coherence scoring in pseudo-labeling during retriever training. To ensure a fair and consistent evaluation, all other factors remain fixed such as the number of retrieved passages (5), the same reader model (SILMA), and the same retriever backbone (bi-encoder model). The retriever models are assessed using key performance metrics, including F1, BLEU-1, ROUGE-L (R-L), and METEOR scores. Bolded values denote the highest scores achieved for each respective metric.

Training Settings	F1	BLEU-1	R-L	METEOR
w/o Coh($\beta = 0$)	19.6	14.6	19.0	15.1
only Coh($\beta = 1, \alpha = 1$)	19.3	14.4	18.8	14.8
only Coh($\beta = 1, \alpha = 0.5$)	19.7	14.9	19.2	15.1
Balanced($\beta = 0.5, \alpha = 0.5$)	**20.5**	**15.4**	**20.0**	**15.9**

TABLE 7. Performance comparison between SNAR Pseudo Labeling and Binary Labels. To ensure a fair and consistent evaluation, the number of retrieved passages is fixed, and the same reader model, SILMA, is employed to comprehend the retrieved passages and generate answers. The table reports key performance metrics, including F1, BLEU-1, ROUGE-L (R-L), and METEOR scores. Bolded values indicate the highest scores achieved for each respective metric.

Labeling Method	F1	BLEU-1	R-L	METEOR
Binary Labels	17.9	13.6	17.5	13.4
SNAR Labeling	**20.5**	**15.4**	**20.0**	**15.9**

When $\beta = 0$ in Equation 3, pseudo-labeling relies solely on semantic relevance, computed using cosine similarity between the question and passage embeddings. This approach yields competitive performance compared to state-of-the-art methods, as discussed in Section IV-D. In contrast, when $\beta = 1$ (rows two and three), pseudo-labeling is based exclusively on narrative coherence scoring, omitting the semantic relevance component. The comparable performance in these settings highlights the significance of narrative coherence scoring in retriever training.

Furthermore, adjusting α in Equation 1 influences the retrieval of broader contextual information. Lowering α encourages the selection of passages that provide supporting context around the central passage, even if they are not directly relevant to the question. This improves performance by capturing a more comprehensive narrative structure.

The optimal results are achieved by integrating both semantic relevance and narrative coherence ($\beta = 0.5$), leading to notable improvements across all performance metrics. These findings indicate that combining these two components enhances the retriever's ability to retrieve contextually relevant passages while maintaining broader coherence and logical flow within narratives. This validates the contribution of narrative coherence scoring in improving the quality of retriever training, reinforcing its importance in training unsupervised retrievers for long narrative texts.

F. ABLATION STUDY III: EVALUATION OF THE PROPOSED NARRATIVE-AWARE TRAINING METHODOLOGY

The results in Table 7 illustrate the benefit of using our SNAR Pseudo Labeling strategy over conventional Binary Labels. In the Binary Labels setting, we employ supervised

contrastive learning where the central passage is treated as the sole positive and all other passages are considered equally negative. This rigid binary distinction, however, overlooks the subtle relevance that some non-central passages may still contribute, especially in narrative contexts.

In contrast, SNAR Pseudo Labeling introduces a soft labeling mechanism where the central passage receives a full relevance score, and all other passages are assigned continuous relevance scores based on their semantic similarity to the question, as defined in Eq. (3). These pseudo-labels are used to train the retriever via regression, enabling it to learn from fine-grained relevance signals.

When evaluating both labeling strategies, SNAR Pseudo Labeling consistently outperforms the Binary Labeling approach across all evaluation metrics. For instance, F1 increases from 17.9 to 20.5 and METEOR from 13.4 to 15.9. These results highlight the importance of modeling nuanced relevance in long narrative texts, where context often spans multiple passages and should not be dismissed by a binary labeling regime.

G. COMPARISON WITH LONG-CONTEXT RETRIEVAL

The experiments delineated in Table 8 assess the efficacy of the proposed SNAR-BI model in comparison to mGTE [57] which outperformed other baselines designed for tasks requiring long-context retrieval [57]. To ensure equitable evaluation, the number of tokens processed by the reader was held constant by altering the number and length of the retrieved passages. Our retrieval framework is initialized with mContriever [27], which is able to process passages comprising a maximum of 512 tokens, in contrast to mGTE [57], which is specifically designed to handle passages of greater length. The findings demonstrate the superior performance of SNAR-BI across all metrics, such as F1, BLEU-1, ROUGE-L, and METEOR.

The SNAR-BI model optimizes the retrieval process by balancing the need for narrative coherence and ensuring the relevance of the retrieved content. This balance is achieved by incorporating our proposed pseudo-labeling mechanism. In contrast, mGTE [57] retrieves excessively long passages, which poses significant challenges for effective retrieval because the mean pooling fails to capture the nuanced semantic distinctions in lengthy passages, which diminishes the precision of similarity computations [49].

H. EFFICIENCY OF THE SNAR RETRIEVER

Table 9 provides a detailed comparison of the performance of GPT4o and SNAR-GPT4o. For GPT4o, the entire narrative book is used as input along with the posed question to generate the answer. In contrast, SNAR-GPT4o utilizes the SNAR-CROSS retriever to identify and select the five most relevant passages (each comprising 400 tokens) from the narrative book, which are then passed to GPT4o. This selective approach ensures that only the most pertinent information is considered while minimizing the inclusion of irrelevant content.

By narrowing the input to the most relevant passages, SNAR-GPT4o demonstrates its efficiency by achieving comparable or even superior performance to the design where the entire book is inputted into GPT4o. This indicates that SNAR-CROSS successfully retrieves the most critical information without losing any important context or details required for accurate answer generation. The ability to maintain, and in some cases enhance, performance highlights SNAR's ability to preserve essential information despite significantly reducing the input size.

Moreover, the results show that SNAR-GPT4o consistently outperforms GPT4o across all evaluation metrics. Notable improvements are observed in BLEU-4 (8.48 vs. 5.99) and METEOR (30.18 vs. 26.25), demonstrating that the retriever effectively directs GPT4o's attention to the most relevant content for each question. These enhancements are particularly valuable in the context of long narrative books, which often exceed GPT4o's maximum token limit of 128,000, necessitating content truncation prior to processing.

In addition, GPT4o, like other language models, faces challenges in attending to information embedded within the middle sections of long inputs, a limitation highlighted in previous studies [8]. SNAR-CROSS addresses this issue by isolating and prioritizing the most relevant passages, ensuring that no critical information is overlooked. As a result, SNAR-GPT4o demonstrates its potential as an efficient solution for narrative QA, particularly for lengthy texts.

The efficiency of SNAR-GPT4o is further evident in the cost savings associated with processing fewer tokens. GPT4o incurs a cost of \$0.00250 per 1,000 tokens .[14] When processing an entire narrative book, which averages 150,000 tokens as shown in Table 1, GPT4o would cost \$0.375 per question. Given GPT4o's maximum token limit of 128,000, even processing the maximum permissible input incurs a cost of \$0.32 per question. In contrast, SNAR-GPT4o selects only five passages (each comprising 400 tokens), reducing the input size to 2,000 tokens per question, resulting in a cost of just \$0.005 per question. This represents a cost reduction of over 98 times. For a batch of 100 questions, the cost using GPT4o alone would be \$37.50, whereas SNAR-GPT4o reduces this to only \$0.50, yielding a savings of \$37.00 for 100 questions. These substantial cost savings, coupled with comparable or superior performance, make SNAR-GPT4o a highly efficient and cost-effective solution for narrative QA.

Another significant advantage of employing SNAR lies in its applicability with other reader models. SNAR effectively identifies and retrieves the most relevant parts of the narrative, leading to (i) improved performance by focusing on critical content and (ii) significantly reduced computational costs due to processing fewer tokens. The SNAR model, with 177 million parameters, is computationally lightweight compared to state-of-the-art reader models, which typically consist of billions of parameters. These larger models are inefficient for processing the entirety of a narrative book

[14]by referring to the pricing page: https://openai.com/api/pricing/

TABLE 8. Performance comparison between SNAR-BI (proposed model) and mGTE. The comparison maintains a fixed number of tokens processed by the reader by adjusting the number and length of retrieved passages, ensuring fair evaluation. Metrics include F1, BLEU-1, ROUGE-L, and METEOR.

Retriever	#Passages	Passage Length	F1	BLEU-1	ROUGE-L	METEOR
mGTE [58]	2	1000	10.7	7.8	10.4	6.6
SNAR-BI	5	400	**20.5**	**15.4**	**20.0**	**15.9**

TABLE 9. A comparative analysis of performance metrics for GPT4o and SNAR-GPT4o is presented. In the case of GPT4o, the entire narrative book serves as the input alongside the posed question to generate the answer. conversely, for SNAR-GPT4o, the SNAR-CROSS retriever identifies and selects the most pertinent passages, which are then concatenated and provided as input to GPT4o along with the question. The table provides F1, BLEU-4, ROUGE-L, and METEOR scores, with the highest values emphasized in bold.

Model	F1	BLEU-4	ROUGE-L	METEOR
GPT4o	27.74	5.99	26.41	26.25
SNAR-GPT4o	**27.87**	**8.48**	**26.64**	**30.18**

due to their high computational resource requirements. By offloading the task of identifying relevant content to SNAR, these reader models can operate more efficiently, enabling practical and scalable solutions for tasks involving long and complex narratives.

I. RETRIEVER SENSITIVITY TO α AND β

To assess the isolated effect of narrative coherence on retriever performance, we conducted an ablation study by varying the hyperparameter α while setting $\beta = 1$ in Equation 3. This configuration ensures that the scoring of non-central passages is governed solely by the exponential distance weight w_i, without influence from semantic similarity. The results are reported in Table 10.

The parameter α regulates how sharply the relevance score decays based on the distance between a passage p_i and the central passage p_{center}. When α is set too small (e.g., $\alpha = 0.0$), the decay becomes negligible, resulting in nearly equal weights for all adjacent passages. This leads the retriever to treat all nearby passages as equally important as the central passage during training. In effect, the model is exposed to a broad set of positive examples that includes both truly relevant and less relevant passages, which dilutes the quality of supervision. This overly soft labeling compromises the retriever's ability to discriminate between core and peripheral content, thereby degrading its retrieval precision. On the other hand, when α is set too large (e.g., $\alpha = 1.0$), the exponential decay becomes overly aggressive. As a result, only the central passage receives a significant score, while even closely adjacent passages are assigned near-zero weights. In this regime, the retriever is effectively trained with a hard positive (central passage) and a hard negative set composed of all surrounding passages. This rigid separation ignores the subtle narrative continuity that exists across adjacent passages, leading to reduced contextual awareness and lower recall. The best performance is achieved at $\alpha = 0.5$, which strikes a balance between these two extremes. At this value, the retriever learns to prefer passages that are closer to the central passage while still incorporating softer relevance signals from its neighbors. This setting promotes a more nuanced supervision signal, enabling the model to leverage the broader narrative structure without introducing noise from distant or weakly related passages.

TABLE 10. Effect of varying α on the SNAR-BI performance. The best results per metric are in bold.

α	F1	BLEU-1	ROUGE-L	METEOR
0.0	10.40	0.076	10.06	0.063
0.1	18.90	14.30	18.40	14.50
0.3	19.55	14.74	18.94	15.02
0.5	**20.50**	**15.40**	**20.00**	**15.90**
0.7	19.26	14.40	18.70	14.80
1.0	19.30	14.40	18.80	14.80

TABLE 11. Effect of varying β on the SNAR-BI performance. The best results per metric are in bold.

β	F1	BLEU-1	ROUGE-L	METEOR
0.9	19.50	14.70	18.90	14.90
0.7	19.90	15.00	19.30	15.30
0.5	**20.50**	**15.40**	**20.00**	**15.90**
0.3	19.90	15.10	19.40	15.60
0.1	19.90	15.05	19.30	15.44

The choice of exponential distance weighting is particularly well-suited for this task, as it provides a smooth, continuous decay of relevance based on distance. Unlike hard windowing or linear scaling, the exponential function naturally emphasizes proximity while allowing for graded inclusion of neighboring context. This reflects the inherent structure of narrative texts, where the most relevant information is typically concentrated around a key event, but surrounding passages often carry supporting details. By modeling this proximity with a flexible decay, the retriever can better capture the coherence of long-form content.

In summary, this ablation confirms that narrative coherence plays a crucial role in effective pseudo-labeling. Carefully tuning α is essential: small values blur the distinction between relevant and irrelevant context, while large values discard useful narrative connections. An intermediate value like $\alpha = 0.5$ facilitates optimal training by aligning the retriever's supervision with the natural flow of narrative information.

The ablation study in Table 11 examines the impact of the hyperparameter β on retriever performance, measured by F1, BLEU-1, ROUGE-L, and METEOR scores. The parameter β regulates the trade-off between narrative coherence and semantic similarity in the scoring function used during

TABLE 12. Performance evaluation of SNAR-BI-SILMA, SNAR-CROSS-SILMA, and SNAR-CROSS-GPT4o models across different question categories. SNAR-BI and SNAR-CROSS are retriever models, while SILMA and GPT4o are reader models. The table highlights challenges in answering implicit, multi-hop, and coreference-dependent questions. The table reports F1, ROUGE-L, and METEOR scores for each category, demonstrating the impact of question complexity on model performance.

Category	SNAR-BI-SILMA			SNAR-CROSS-SILMA			SNAR-CROSS-GPT4o		
	F1	ROUGE-L	METEOR	F1	ROUGE-L	METEOR	F1	ROUGE-L	METEOR
Implicit/Explicit									
Explicit	25.08	24.61	20.85	27.19	26.60	22.89	32.99	31.91	35.75
Implicit	16.10	15.61	11.20	16.63	16.10	12.10	22.99	21.61	24.89
Multi-Hop									
Single-Hop	22.44	21.99	18.30	24.39	23.80	20.22	30.54	29.31	33.33
Multi-Hop	16.17	15.64	10.69	16.07	15.61	11.14	22.08	20.83	23.33
Coreference									
Without Coreference	22.55	22.04	17.97	24.47	23.92	20.14	30.63	29.50	33.36
With Coreference	16.52	16.12	11.99	16.66	16.10	12.08	22.65	21.21	24.17
Question Intent									
Action	17.61	17.19	13.75	20.71	20.30	17.03	25.50	24.39	27.28
Causal Relationship	19.82	19.33	15.69	20.85	20.06	16.56	26.23	24.70	30.18
Character	28.38	28.11	22.76	26.38	26.13	21.70	35.00	34.23	37.54
Feeling	13.09	12.73	9.35	15.47	15.12	11.62	23.08	21.95	23.24
Outcome Resolution	22.46	21.60	17.66	23.24	22.89	18.32	27.31	25.65	29.11
Prediction	18.40	17.58	12.97	14.76	14.27	11.50	21.94	20.42	25.63
Setting	28.12	27.66	21.06	35.66	34.84	28.48	41.29	40.52	41.36

retriever training, where higher values emphasize narrative coherence and lower values favor semantic similarity.

The results indicate that a balanced setting of $\beta = 0.5$ achieves the best performance across all evaluation metrics, with the highest F1 (20.5), BLEU-1 (15.4), ROUGE-L (20.0), and METEOR (15.9) scores. This suggests that optimal retriever training is achieved when both narrative structure and semantic relevance are jointly considered. In contrast, skewing the scoring toward either extreme using high values (e.g., $\beta = 0.9$) or low values (e.g., $\beta = 0.1$) results in diminished performance, underscoring the necessity of balance.

These findings emphasize the importance of incorporating both global narrative coherence and local semantic similarity when training retrievers for long-form narrative QA tasks.

J. PERFORMANCE ANALYSIS

Table 12 shows the performance evaluation of SNAR-BI-SILMA, SNAR-CROSS-SILMA, and SNAR-CROSS-GPT4o models across different question categories. SNAR-BI and SNAR-CROSS are our retriever models, while SILMA and GPT4o are reader models. The results reveal consistent declines in performance when questions necessitate more complex reasoning. All models exhibit higher scores on explicit questions compared to implicit ones, underscoring the relative simplicity in locating information directly stated in the text. Similarly, single-hop questions—answerable within a single paragraph—evoke stronger performance than multi-hop questions, which require the synthesis of details dispersed across multiple paragraphs. A similar trend is observed with coreference, where queries not requiring entity tracking throughout the text achieve higher scores compared to those that do involve such tracking. Notably, the GPT4o reader shows a significant advancement over the SILMA model across all categories, attributable to its superior reasoning and contextual understanding capabilities.

Furthermore, examining the Intent category provides additional insights into how models perform when confronted with various narrative question types. In particular, the Setting subcategory, which focuses on inquiries pertaining to time, location, and broader contextual details, consistently shows robust performance across all evaluated models. This is evident in questions such as "What is the place that attracts American tourists in the story?"(ما هو الحادث الذي وقع في بارادايس؟) and "What is the incident that occurred in Paradise?"(ما العلاقة بين جون بريک وماري بيوري؟). These examples require identifying concrete, explicit details within the narrative, thereby facilitating accurate retrieval and synthesis. A similar trend is observed for character-focused questions, which often probe roles, relationships, and identities. For instance, the question "What is the relationship between John Break and Mary Beury?"(من هو الشخص الذي قابل برادن في العيادة؟) highlights a marital relationship either explicitly stated or clearly implied by the text. Likewise, a question such as "Who is the person that Braden met at the clinic?"(كيف كانت ماري بيوري تشعر تجاه السيد برايس؟) relies on the model's ability to map named entities to specific events, thus minimizing ambiguity. Because such questions typically reference a single, well-defined scenario within the narrative, they allow the model to generate accurate answers.

On the other hand, the Feeling and Prediction categories pose notable challenges for many models. For instance, a Feeling question such as "How did Mary Beury feel about Mr. Price?" (ما الذي كان يخطط له ثورندايک في اليوم التالي؟), which may demand an answer like "She despises him." (تكرهه بشدة.), requires interpreting the emotional states of

characters based on subtle textual cues. These cues, which can be implied rather than explicit, make it difficult for models to infer the intensity or nuance of a character's feelings. Similarly, Prediction questions—such as "What was Thorndike planning for the next day?" (ما الذي كان يخطط له ثورندايك في اليوم التالي؟)—demand insight into possible future actions or outcomes, often relying on indirect evidence or character motivations. These findings highlight the difficulty in addressing questions that require more sophisticated reasoning capabilities, where even advanced models like GPT4o exhibit notable gaps.

Section IV-H demonstrated that providing GPT-4o with the output generated by our retriever yielded performance comparable to, and in many cases exceeding, that of directly inputting the entire book into the model. This result underscores the retriever's effectiveness in isolating the most relevant and contextually rich segments necessary for accurate answer generation, while preserving the essential narrative context. Moreover, by processing only a subset of the book's tokens rather than the complete text, this approach achieved a significant reduction in computational cost, reducing expenses by a factor of 98 compared to the full-book input methodology. These findings highlight the efficiency and scalability of our retrieval-augmented framework. The majority of the residual errors are primarily attributed to limitations in the reader model, which are discussed in detail in the following analysis:

Count reasoning errors occur when the model fails to accurately identify and enumerate entities within the context. For instance, in response to the question "How many servants are in Mr. Mander's apartment?" (كم عدد الخدم في شقة السيد ماندر؟), the model must extract and count six individuals: a butler, a personal servant, a cook, two maids, and a parlor maid. Incorrect responses typically result from omissions or miscounts.

Multi-hop reasoning limitations represent another major source of failure. Narrative QA often requires synthesizing information from multiple parts of the text. For example, the question "How did Thorndyke's attempt to open the sealed room end?" (كيف انتهت محاولة ثورندايك لفتح الغرفة المنيعة؟) demands tracking a sequence of events to infer that Shymonds was fatally trapped due to the rebound mechanism in the lock. Many models struggle with this form of compositional reasoning across distributed narrative elements.

Misinterpretation of question intent leads the model to focus on irrelevant information. For example, in the question "How did Thorndyke discover Creek's body?" (كيف اكتشف ثورندايك جثة كريك؟), the model may incorrectly respond that "James Brody discovered the body," overlooking the correct detail that Thorndyke used a flashlight to reveal the body. Such mistakes indicate a failure to align the model's output with the actual information requested in the question.

Inference of implicit emotional states presents a more subtle challenge. Narrative texts often express emotions indirectly, and current models tend to miss such nuances. Consider the question "How did Wang Lung feel when he considered selling the child?" (كيف شعر وانج لنج عندما فكر في بيع الطفلة؟). The model predicted "Sad" (حزن), which partially captures the emotional tone but lacks the specificity of the ground truth: "He felt anxious and hesitant" (شعر بالقلق والتردد ولم يحتمل الفكرة).

Hallucination from inadequate retrieval is another notable issue, arising when the retriever fails to locate sufficient or relevant content. In such cases, the reader model tends to generate speculative answers that are not supported by the available context, leading to responses that diverge significantly from the narrative truth.

The findings underscore the pivotal role of employing a robust reader model to significantly enhance overall performance, particularly in complex reasoning tasks. While the underlying retriever demonstrates effective performance within the RAG framework, the results reveal notable limitations in the reader model's ability to fully comprehend and leverage the retrieved passages to generate accurate responses, especially for questions that require implicit meanings, multi-hop reasoning, and coreference resolution. These challenges highlight the need for future research to focus on the development and training of advanced reader models, specifically designed to address the complexities of narrative texts and ensure deeper contextual understanding for improved QA outcomes.

V. LIMITATIONS

In the process of pseudo-labeling, as described in equation 3, the semantic relevance of each passage to the corresponding question is determined by calculating the cosine similarity 2 between the question's embedding and the passage's embedding. These embeddings are generated using the openai-small embedding model. This methodology could be enhanced by employing more sophisticated embedding models, such as the openai-large model. Additionally, a notable limitation of this work is its focus on the retriever component; future research endeavors could explore the development of advanced reader models, specifically designed to address the complexities inherent in narrative texts.

While synthetic data generated by LLMs can be valuable, particularly when real data is limited, it carries inherent limitations, notably the risk of introducing biases [58], [59]. Although our framework utilizes LLMs to generate semantically rich question-passage pairs, these models may inadvertently skew the diversity of the resulting data. For example, LLM-generated questions often over-represent specific linguistic patterns or question types, which may lead the retriever to favor certain phrasings while neglecting a broader range of question formulations.

This bias can subsequently limit the retriever's generalizability to naturally written questions. Addressing this challenge requires future exploration into strategies that enhance prompt diversity, such as explicitly instructing the LLM to generate more challenging and varied types of questions.

VI. CONCLUSION

In this paper, we addressed the critical gap in Arabic QA for lengthy narrative books by introducing AQANB, a dedicated dataset designed to evaluate QA models on extended textual contexts. To enhance retrieval accuracy in narrative QA, we proposed an annotation-free retriever training framework that utilizes LLMs to generate synthetic question-passage pairs, eliminating the need for costly manual annotations. Our approach incorporates a novel narrative coherence-based scoring mechanism, ensuring that relevant passages and their surrounding context are effectively captured to preserve narrative coherence. The experimental results on AQANB demonstrated the superiority of our method, achieving significant performance gains over commercial and open-source retrievers. These findings highlight the potential of unsupervised retriever training in improving passage retrieval for long-form QA. As this work focuses on the retriever part, future research can the development of advanced reader models, specifically designed to address the complexities of narrative texts.

VII. ETHICAL CONSIDERATIONS

In the data collection phase, each contributor ensured that the stories were appropriate within Arab culture and do not include inappropriate, potentially harmful, or culturally sensitive information. The stories were exported from Hindawi.org is a non-profit organization dedicated to promoting knowledge and culture among Arabic speakers. According to its privacy policy, the platform provides free access to its content, making it an invaluable resource for educational and learning purposes.

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